

Research paper

Numerical investigation of window design strategies for energy efficiency and thermal–visual comfort in buildings

Tahmid Talukder^a, Md. Mahmodul Alam^{a,*}, Pronab Debnath^a, MAH Shovon^b, Fahim Tanvir^a, Nazmush Sakib^a, Ahbab Faiyaj Haque^a, Md. Hasan Ali^a

^a Department of Energy Science and Engineering, Khulna University of Engineering & Technology, Khulna 9203, Bangladesh

^b Department of Civil Engineering, Khulna University of Engineering & Technology, Khulna 9203, Bangladesh

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ABSTRACT

Window design plays a decisive role in energy performance, daylighting, and occupants' comfort in buildings. This work investigates the effects of window shape and glazing type on energy consumption as well as indoor thermal and visual comfort in a five-story institutional building under the hot and humid climate of Bangladesh. The study compares twenty configurations, combining four window shapes (rectangular, arched, Palladian and bay) with five single-glazed tinted glass types (clear, blue, bronze, green, and grey; all 6 mm). Annual simulations were performed in DesignBuilder using the EnergyPlus engine to assess energy consumption, cooling demand, lighting load, solar heat gains, daylight availability, and comfort indicators such as Predicted Mean Vote (PMV), percentage of thermal, and visual comfort hours. Rectangular clear configuration was taken as the baseline. Results show that the Palladian green glass configuration offered the best performance, reducing annual energy use by 8.56% compared to the baseline, with a monthly average PMV of 1.88 and thermal comfort maintained for 32.1% of occupied hours. Bay clear windows showed the poorest performance with the highest energy use. Although the Palladian green configuration reduced cooling demand and increased thermal performance, the glass was darker and yielded a lower daylight, reducing visual comfort as compared to the clear glazing. In general, the results from this study suggest that the green glazing incorporated with the Palladian windows can contribute positively to reducing building energy demand and thermal comfort in hot and humid climates, although complementary solutions could be needed to fulfill acceptable visual comfort.

1. Introduction

The growing global energy demand for increasing population and industrialization, requires urgent focus in energy management and efficiency, particularly within the built environment [1,2]. In most countries, buildings account for approximately 40% of total energy consumption, which places a lot of pressure on the world resources and is a major cause of climate change [3]. In response, international policies mandate aggressive energy reduction targets, such as achieving “nearly zero” energy consumption in new buildings [4]. Ensuring sustainable and decarbonized building operation requires significant efforts toward identifying and implementing economically viable technologies and solutions [5,6].

The building envelope, particularly openings in buildings plays a crucial role in ventilation, daylighting, and thermal regulation, directly influencing energy consumption, indoor comfort, and air quality [1,7].

However, the improvement of natural illuminance and visual connection with the outdoor environment often directly conflicts with the reduction of heating, ventilation and air conditioning (HVAC) loads [8,9]. As a result, different window design strategies are required to optimize both energy efficiency and Indoor Environmental Quality (IEQ) [1,10].

In contemporary building energy research, single glazing is often examined as a baseline configuration to highlight the performance improvements offered by advanced fenestration systems. Ahmed et al. [11] explored the impact of Window-to-Wall Ratio (WWR) and orientation on the heating and cooling energy of an office building in Kirkuk, Iraq, using single clear and gray glass as benchmark configurations. The researchers reported that for single clear glass, the best-case cooling energy was 1406 kWh with a 25% WWR facing north, while the minimum heating energy was 72 kWh with a 50% WWR facing south. Conversely, the worst-case scenario involved a 100% WWR south-facing single clear

* Corresponding author.

E-mail address: mahmod@ese.kuet.ac.bd (Md.M. Alam).

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window, which consumed 3602 kWh for cooling. El Krimi et al. [12] conducted dynamic thermal simulations in Python to quantify the indoor thermal behavior of apartments equipped with standard single glazing ($U = 5.2 \text{ W/m}^2\text{K}$) across six Moroccan cities. The results revealed that single glazing exhibited the greatest thermal instability and the lowest comfort duration among all glazing types tested. However, the authors concluded that single glazing is generally unsuitable for extreme climates due to wide temperature fluctuations and weak thermal stability, often resulting in indoor peaks reaching 40–48 °C during summer if not properly oriented or shaded. Al-Harbi et al. [13] utilized a conjugate numerical analysis with ANSYS Fluent to investigate solar heat gain through single-pane clear and tinted glass in Riyadh, Saudi Arabia. The study established that single-pane clear glass (6 mm) resulted in the highest total heat gain of 611.3 W/m^2 at an air change rate (ACH) of 5, which was dominated by solar transmission gain (554.7 W/m^2) rather than conduction (56.6 W/m^2). Although single-pane tinted glass reduced the solar transmission gain to 297.2 W/m^2 , it simultaneously increased the glass conduction heat gain to 107.3 W/m^2 due to elevated glass temperatures.

On the other hand, research focuses on advanced glazing and façade systems designed to harness or reject solar radiation dynamically, moving beyond conventional high performance double-glazed or low-emissivity (LoE) windows [14,15]. The development of smart windows with dynamic and adaptive glazing offers a critical avenue for balancing energy conservation and occupant comfort [16]. Shi et al. [17] evaluated the annual performance of TC glazing, focusing on thermal and visual comfort. On the other hand, enhancing energy efficiency should not compromise IEQ. As the factors of IEQ i.e. thermal comfort, visual comfort, acoustic comfort, indoor air quality (IAQ), directly influence occupant health and productivity [18]. Thermal comfort is commonly quantified using the Predicted Mean Vote (PMV) and Predicted Percentage of Dissatisfied (PPD) indices, calculated based on internal and external factors [3,10,19]. The use of external shading devices is essential to minimize overheating and prevent excessive glare [7,10,20]. Innovative façade geometries, such as the multi-angled façade systems studied by Hannoudi et al. [20], aim to optimize solar exposure and daylighting in relation to varying orientations. Alsukkar et al. [7] investigated parametric split louvers for enhanced daylighting performance and reduced lighting energy consumption.

Integrating Phase Change Materials (PCMs) into glazing units has garnered significant attention due to their ability to store latent heat, thereby enhancing the thermal mass and delaying peak cooling/heating loads [2,21]. Hu et al. [22] further explored the optimization of thermal comfort and daylight performance using PCM glazing.

Not only the glass materials but also the geometric shape of the façade has a crucial impact on building energy consumption. The geometry of windows affects solar exposure, incident radiation angles, and the distribution of daylight within indoor spaces. Recent studies have shown that different window layouts and geometrical configurations can significantly alter daylight distribution patterns, illuminance levels, and energy performance [23]. WWR, fenestration orientation and shading devices are the key parameters that affect building energy consumption [6,8,24]. The WWR strongly affects heating, cooling, and lighting demands [6,25]. Cherier et al. [26] provided a detailed parametric analysis demonstrating how the interaction of glazing type, WWR, and orientation determines optimal energy savings across various Algerian climates and found the suitable WWR of 60%, except for a few cases. Gigasari et al. [24] emphasized that optimal WWR selection is highly dependent on climate and orientation. Using DaySim 3.2, Acosta et al. [27] analyzed square, horizontal, and vertical window forms and observed that horizontal configurations improve uniformity of illumination, while elevated window positions increase rear-zone daylight autonomy by approximately 30%. Similarly, Do et al. [23], through Ladybug simulations, compared various window geometries and concluded that rectangular openings enable greater daylight penetration than square ones, with horizontal layouts achieving superior spatial daylight

autonomy compared to vertical designs. Therefore, evaluating different window geometries is essential to understand their combined impact on thermal and visual performance in buildings.

Visual comfort relates to adequate daylight availability, typically quantified using metrics like Useful Daylight illuminance (UDI) [7]. Furthermore, the quality of the window view significantly affects occupant satisfaction, cognitive performance, and emotional state [19, 28]. Ko et al. [9] reported that geometric variables, such as glazing area and distance, significantly correlate with occupants' satisfaction with the view. The optimal design often requires solving complex optimization problems (MOGA, NSGA-II) that minimize energy consumption while maximizing daylight performance and comfort ratios [6].

Advanced glazing technologies have shown strong potential for improving building energy performance and indoor comfort [14,16]. However, their use is still limited in many developing countries due to high cost, technical complexity, and maintenance requirements. In contrast, single-glazed tinted glass is widely used in institutional buildings in countries like Bangladesh because it is affordable and easy to install. Therefore, it is important to evaluate the performance of these commonly used glazing systems under local climatic conditions to develop practical and applicable design strategies.

Although extensive research is available for building energy conservation, and thermal comfort and daylight, significant gap remains with respect to the overall comparative assessment of window shapes and glass types particularly for institutional buildings in hot and humid climate like Bangladesh. Most previous studies have primarily focused on isolated parameters such as window-to-wall ratio, advanced glazing technologies, or shading devices. Other studies looked only at different types of buildings. Currently, there are a few comprehensive studies on the combined effects of window shapes and conventional tinted glazing on building performance. In particular, the interaction between window shape and solar-optical properties of tinted glass remains underexplored, especially in cooling-dominated climates. This gap is significant because window geometry can alter solar exposure and daylight distribution, while tinted glazing modifies solar heat gain and visible transmittance, leading to complex performance trade-offs. This study examines how those factors impact cooling energy and lighting consumption. And also look at indoor conditions, such as thermal and visual comfort. Secondly, the distinct climatic conditions and building culture in Bangladesh suggest that local research is necessary to develop situation-specific window design guidelines. The limitations of current literature and the corresponding research contributions of this study are detailed in Table 1.

The novelty of this work lies in its integrated approach of window shape and glazing types to identify the optimal window design on energy consumption of an institutional building. This study provides a systematic comparative assessment of the thermal and visual comfort as well as reducing the energy consumption in a hot and humid climate of Bangladesh. To address the study aims, a five-story institutional building was modeled and simulated under the climatic conditions of Bangladesh, where various window shapes and glazing types were systematically evaluated in terms of energy consumption, thermal comfort, and visual comfort.

2. Methodology

This section details the systematic approach employed in the development, simulation, and evaluation of the building energy model. It encompasses model construction, material and configuration selection, simulation parameters, scenario development, and the data collection and analysis techniques necessary to fulfill the research objectives.

2.1. Research framework

Fig. 1 shows the complete workflow of the simulation to investigate

Table 1

Contribution of current study in the context of research gap.

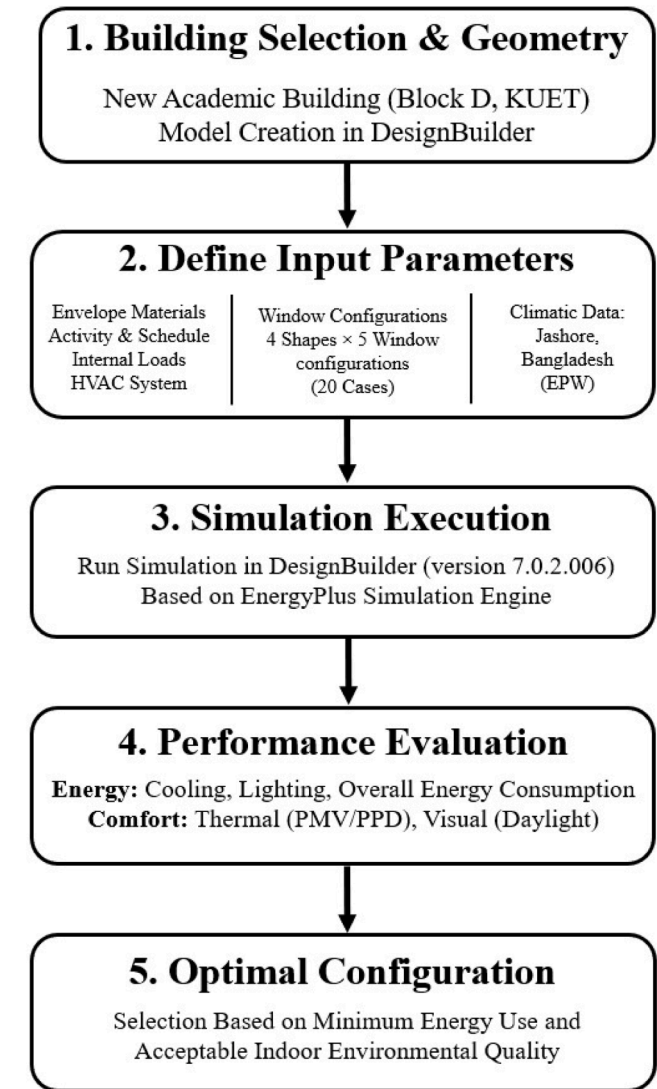
Area of Investigation	Literature Gap (Prior Work)	Research Contribution (This Study)
Advanced Glazing Systems	Focus often limited to single technologies (e.g., PCM [6,22] or TC [17,29]) lacking comprehensive comparisons with simple tinted glazing types [15,25,30].	Systematically compares multiple traditional (tinted glass) against baseline clear glazing, providing a direct metric for evaluating the effectiveness of basic spectral selectivity.
Geometric Design Parameters	Optimization primarily focuses on WWR [24] or simple shading devices [7,10]. The impact of window shapes (e.g., Palladian, arched, bay) is rarely assessed relative to energy efficiency and IEQ metrics [10].	Investigates the integrated effects of four distinct window shapes (rectangular, arched, Palladian, bay) and their interaction with glazing properties on overall building performance
Multi-Objective Optimization (MOO)	MOO studies prioritize standard material properties (U-value, SHGC) [6] or operational aspects. Few studies integrate shape, tint properties, and IEQ comprehensively for optimization in specific challenging climates [10].	Provides a comparative assessment across twenty distinct configurations (shape * tint) to identify the global optimal combination based on reduced energy use, improved thermal comfort (PMV), and daylight availability
Climate Specificity and Applicability	Generalization of findings across diverse climates is challenging [24]. Hot-humid climates present unique cooling load challenges due to high solar gain and humidity [6,21].	Conducts an annual simulation to justify the seasonal variation specifically for the hot and humid climate of Bangladesh, which is relevant for similar cooling dominant regions
IEQ Assessment	Studies often focus on energy reduction, neglecting IEQ factors, particularly the trade-off between thermal and visual comfort [9].	Balances the energy efficiency considering both thermal comfort and visual comfort.

the effect of using different window configurations on the energy consumption and indoor thermal and visual comfort. The simulations were executed using DesignBuilder (version 7.0.2.006). Fig. 1 illustrates the overall methodological framework adopted in this study. The process starts with the building selection and geometric model development of the selected building in DesignBuilder software. Subsequently, the key input parameters such as, building envelope materials, occupancy schedule, internal loads, HVAC system characteristics, location specific climate data and different window configurations are defined. After executing the simulations, the process is followed by a performance analysis focusing on energy use and indoor comfort, leading to the identification of the configuration that offers the best balance between efficiency and environmental quality.

2.2. Window shape and glass type

Window is the principal element for reducing energy consumption and impact on achieving significant amount of daylighting penetration [31]. Four different window shapes with combination of five different types of single-glazed glasses with uniform thickness of 6 mm were examined.

The rectangular window configuration was used as the baseline case, with a window-to-wall ratio (WWR) of 30%. For other window shapes (arched, Palladian, and bay), the overall height and width were kept identical to the baseline rectangular window to ensure consistent façade proportions. However, due to geometric differences, the effective glazed area varied among the window types, resulting in slight variations in WWR across the cases. These variations reflect realistic architectural conditions, where changes in window shape inherently influence the

**Fig. 1.** Workflow of the simulation process.

glazing area.

The measurements of each window type along with their shape are illustrated in Fig. 2. All the measurements are in meter. The areas were 3.36 m², 2.83 m², 2.70 m² and 4.256 m² for Rectangular, Arched, Palladian, and Bay respectively.

For each shape of windows, five different single-glazed glass types with a uniform thickness of 6 mm were used, including clear, blue, green, bronze and grey variants. Properties like, U-value, thermal conductivity and thermal emissivity were same for every type of glasses. However, the Solar Heat Gain Coefficient (SHGC) and Visible Transmittance (VT) differed, as presented in Table 7, leading to variation in both solar heat gain and daylight penetration. A total of 20 simulation scenarios were conducted, representing all combinations of the four window configurations and five glass types. Each scenario was simulated under identical building geometry, internal loads, and climate conditions to ensure a fair comparison. The outputs of energy consumption, thermal comfort, and daylighting from each case were then collected for detailed analysis.

2.3. Building model description

The New Academic Building of Khulna University of Engineering and Technology (KUET), located in Bangladesh, has been chosen as the case

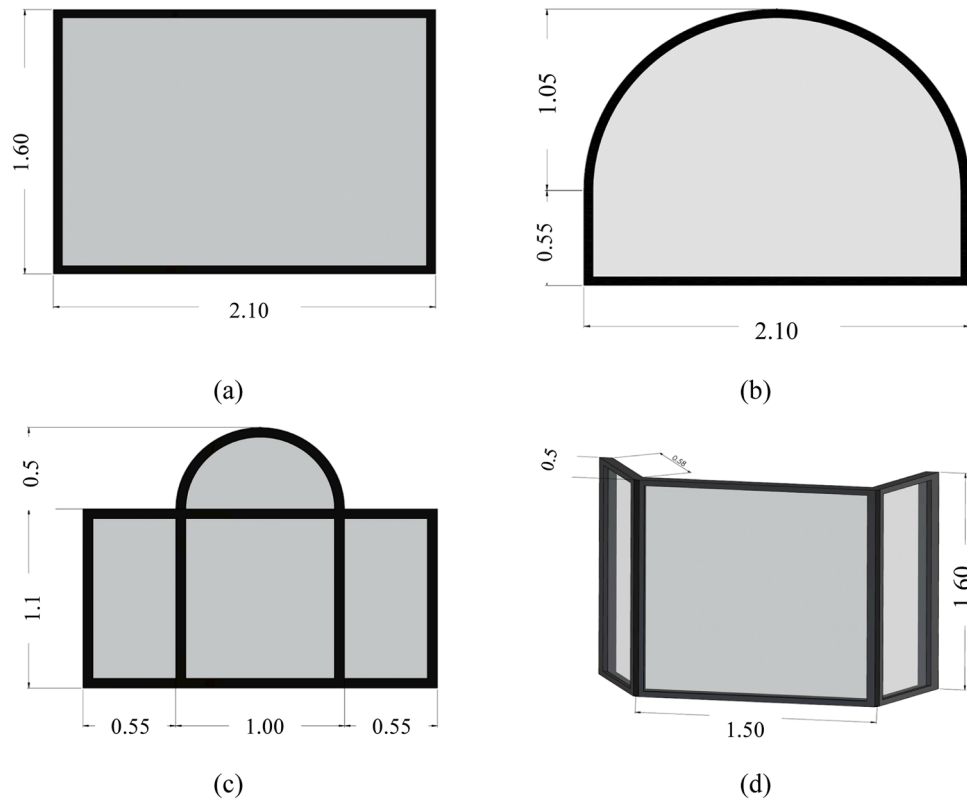


Fig. 2. Measurement of different shaped windows, (a) Rectangular, (b) Arched, (c) Palladian, (d) Bay.

study building. It is a five-story building with different thermal zones, including classrooms, laboratories, faculty offices, and public circulation spaces. The floor area of this building is approximately 4920 m². The extensive fenestration of this building presents a compelling basis for its selection as a case study building. Furthermore, its strategic location supports the justification for choosing it. So, analyzing the result can easily highlight the change in energy consumption pattern and indoor thermal and visual comfort for different window design strategies under hot and humid climate. Fig. 3(a) depicts the building model with sun path projection and Fig. 3(b) illustrates the position of the building on map.

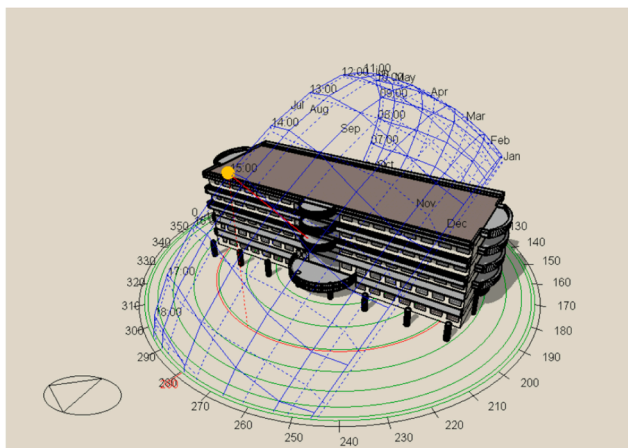
The study building is located in Khulna, southwestern Bangladesh, which experiences a hot and humid climate. For simulation purposes,

2023 climate data from the nearby city of Jashore (23.18°N, 89.17°E; “BGD_KH_Jessore.AP.419360_TMYx.2023”) were employed, representing ASHRAE Climate Zone 1A with time zone of GMT+06:00.

In the eastern side of the building there are C block, B block and A block respectively. But the potential shading from these blocks is not considered in this study to determine the actual amounts of solar gains on external windows and the effect of the glazing materials. Table 2 shows the total height, area and all other parameters of the selected building.

2.4. Floor plan

The internal zoning for each floor was done following the layout of



(a)



(b)

Fig. 3. Photograph of new academic building (D Block), a) Building model with sun path projection, b) Location map.

Table 2

Description of new academic building (D Block).

Parameters	Description
Building Type	Institutional Building
Location	Khulna, Bangladesh
Orientation	Main façade facing East–West
Number of Floors	5 Stories
Floor Area	4920.01 m ²
Building Footprint	55.25 m*17.81 m
Floor-to-Floor Height	3.5 m
Total Height	17.5 m
Window-to-Wall Ratio (WWR)	30%

the building. All functional zones within the building are identified based on the number of structural spans that the zones occupy. The size of each span is 7 m × 3 m. Table 3 shows the types of spaces across the floors, along with their schedules and the number of spans they occupy. Using these data, the 3D model of the case study building was created in DesignBuilder software.

2.5. Building envelope constructional data

Heat transfer from outdoor environment to indoor environment depends on the properties of the construction materials of the envelopes of the building [32]. The New Academic Building (D Block) has external wall, internal partition, roof and floor as building envelope. For the simulation, construction details of the building envelope for each floor are required. The construction data, including the material layers and U-values for each type of envelope, are shown in Table 4.

2.6. Occupancy, lighting and equipment load

Variations in occupancy, lighting, and equipment usage significantly influence the thermal behavior and energy performance of a building [33,34]. These parameters are defined as internal loads, and they are essential for accurately estimating electrical energy demand and comfort parameters. Each zone has a unique activity schedule. Depending on its function, the operating hours are also unique for each zone. Table 5 illustrates the schedules (Table 3) of different zones along with occupancy density, lighting power density, and equipment power density.

2.7. HVAC and lighting system

The parameters of the HVAC system and lighting system play important roles in this simulation, since they account for significant amount of the overall building's energy consumption [35]. The various parameters of the HVAC system as well as lighting conditions of the structure being investigated are presented in Table 6. The HVAC system used in the building consisted of split-type air conditioners without a fresh air supply. To enhance the accuracy of the simulation, the HVAC systems were assigned to only those zones where they actually exist in

the actual building. Both humidity control and natural ventilation strategies were implemented in the simulation. Lighting control was incorporated in the simulation setting to automatically switch on and off the artificial lights depending on the illuminance level, ensuring energy efficient operation.

2.8. Simulation settings and procedure

Standard DesignBuilder simulations were applied uniformly across all scenarios to ensure consistent and comparable results. To capture the seasonal variations, every simulation case was run for full year (8760 h), with hourly data extracted for detailed analysis of energy consumption, thermal comfort, and daylighting. Lighting control was enabled to reduce lighting energy use; when daylight was sufficient, artificial lighting remained off. Fig. 4 represents all the cases of the simulations.

2.9. Performance evaluation and selection criteria

2.9.1. Energy performance

Energy performance was assessed based on annual simulation results, including total energy use, cooling demand, and lighting consumption. Cooling energy primarily represents the influence of solar heat gains through window systems, whereas lighting energy is governed by the availability of daylight within the indoor environment. These metrics were used to evaluate the overall energy efficiency of various window configurations.

2.9.2. Thermal comfort performance

Thermal comfort performance was analyzed through the PMV and PPD indices following the Fanger model framework. These indices were computed using hourly simulation results, accounting for environmental and personal factors such as air temperature, mean radiant temperature, air velocity, humidity, clothing insulation, and metabolic activity. In parallel, the percentage of comfort hours over the year for the occupied hours was estimated based on operative temperature to evaluate the duration of favorable indoor thermal environments.

2.9.3. Visual comfort performance

Visual comfort was evaluated using multiple daylighting indicators, including indoor illuminance levels (lux), daylight factor (DF), percentage of visual comfort hours, and excessive glare occurrence. Hourly simulation data were used to classify indoor conditions into comfortable, under-lit, and glare conditions based on standard illuminance thresholds for educational spaces. According to the study conducted by Yong et al. [36], it is recommended that the illuminance level on reading planes in classrooms designed for students with low vision should range from 276.67 lux to 701.59 lux. In this work, the illuminance below 250 lux is considered as under-lit place and illuminance level over 700 lux is considered as excessive glare region.

A comparative analysis of all scenarios was conducted to identify the optimal window configuration, considering the trade-offs among energy

Table 3

Floor-wise allocation of various functional zones of the new academic building (D Block) used in the energy simulation model.

Space Type	Number of Span	Schedule	Count				
			Ground Floor	1st Floor	2nd Floor	3rd Floor	4th Floor
Classrooms	4	Class Schedule	4	0	4	2	4
Faculty Rooms	1	Block Schedule	0	12	8	0	5
Office & Consulting Area	2	Block Schedule	0	1	1	0	1
Laboratories	4	LAB Schedule	3	2	6	2	2
Library	2	Library Schedule	1	0	0	0	0
Seminar Hall	5	Seminar Schedule	0	1	1	0	0
Toilet	1	Toilet Schedule	2	2	2	2	2
Stairs	1	Stairs Schedule	1	1	1	1	1
Lift	1	Lift Schedule	1	1	1	1	1
Open Space	-	Toilet Schedule	1	1	1	1	1

Table 4
Building envelope construction specifications and thermal performance.

Construction Type	Description	Specification	Cross Section
External wall construction (base wall)	Layers (Outermost to Innermost)	12.5 mm Cement Plaster 229 mm Brickwork Inner 12.5 mm Cement Plaster	
	U-value	1.742 W/m ² ·K	
Roof construction (Baseroof)	Layers (Outermost to Innermost)	38 mm Stone Chippings for Roofs 127 mm Concrete, Reinforced (with 1% steel) 6 mm Cement Plaster	
	U-value	4.113 W/m ² ·K	
Internal Partition Construction	Layers (Outermost to Innermost)	12.5 mm Cement Plaster 125 mm Brickwork (Inner) 12.5 mm Cement Plaster	
	U-value	2.015 W/m ² ·K	

Table 5
Schedule details of different functional spaces in D Block, including operating hours, occupancy density, lighting power density, and equipment power density.

Schedule Name	Operating Hours	Occupancy Density (people/m ²)	Lighting Power Density (W/m ²)	Equipment Power Density (W/m ²)	Day/Week
Class Schedule	08:00–13:00	0.3671	4.27	12.3	5
Block Schedule	08:00–13:00, 14:30–17:00	0.0952	3.91	16.88	5
LAB Schedule	14:30–17:00	0.1063	4.77	17.14	5
Library Schedule	12:00–13:00	0.1426	4.15	9.33	3
Seminar Schedule	12:00–13:00	0.1063	4.15	17.14	1
Toilet Schedule	08:00–17:00	0.0829	3.28	N/A	7
Stairs Schedule	08:00–17:00	0.0829	2.74	N/A	7
Lift Schedule	08:00–17:00	0.4861	3.26	3.5	7

Table 6
HVAC parameters and lighting specification.

HVAC System	
System Type	Split No Fresh Air
CoP	2.930
Cooling	24.0 °C
Cooling Setback	28.0 °C
Heating	18.0 °C
Heating Setback	12.0 °C
Humidification setpoint	10%
Dehumidification setpoint	90%
Natural Ventilation	Indoor mean temperature 24.0 °C
Lighting System	
Lighting Display Density	4.77 W/m ²
Target Illuminance	250 lux
Lighting Control	On

performance, thermal comfort, and visual comfort indicators. Configurations with reduced energy demand and satisfactory indoor environmental quality were prioritized. This systematic approach supports the determination of a well-balanced and feasible window design solution.

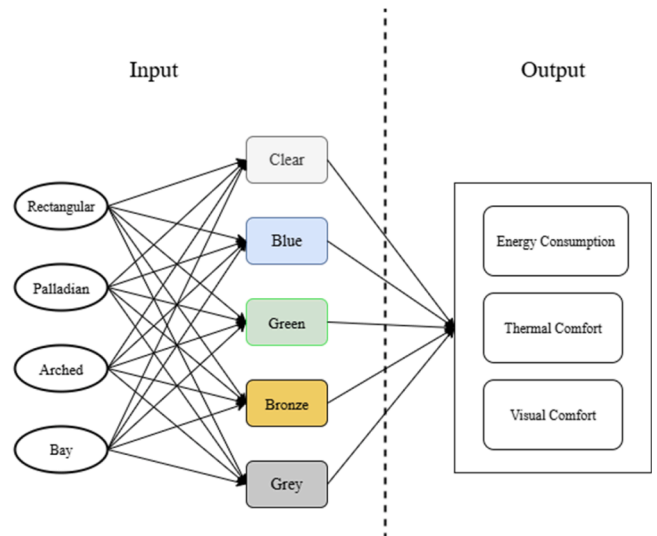


Fig. 4. Overview of the simulation cases illustrating the combination.

Table 7

Thermal and optical property of the glazing types.

Pane Type	Solar Heat Gain Coefficient (SHGC)	Visible Transmittance (VT)
Generic Clear (Existing)	0.819	0.881
Generic Blue	0.62	0.57
Generic Green	0.623	0.749
Generic Bronze	0.62	0.534
Generic Grey	0.602	0.431

2.10. Mathematical formulation for parameters calculation

2.10.1. Solar heat gains calculation procedure

If conduction heat gain or heat loss in the absence of solar radiation,

$$\frac{\dot{Q}_{cond}}{A} = U(T_0 - T_i) \quad (1)$$

If solar radiation strikes on the glazing, which is transmitted and absorbed by the glazing,

$$\frac{\dot{Q}_{solar, gain}}{A} = SHGC \cdot I_T \quad (2)$$

where I_T represents the solar irradiance on the glazing surface. SHGC is the ratio of solar heat gain via the window to the solar radiation incident upon the window.

$$SHGC = \frac{\text{Solar heat gain through the window}}{\text{Solar incident radiation on the window}} = \frac{\dot{q}_{solar, gain}}{\dot{q}_{solar, incident}} \quad (3)$$

The more accurate and operational version of Eq. (8) is,

$$\frac{\dot{Q}}{A} = SHGC_{dir} \cdot I_{T, dir} + SHGC_{dif} \cdot I_{T, dif} \quad (4)$$

In the presence of solar radiation, the total heat gain becomes,

$$\frac{\dot{Q}}{A} = \frac{\dot{Q}_{cond}}{A} + \frac{\dot{Q}_{solar, gain}}{A} = U(T_0 - T_i) + \tau I_t + h_o \Delta T_{sol} \quad (5)$$

To simplify the equation, ΔT_{sol} is replaced with I_t , yielding a linear relation independent of T_i and T_o [37]. Performing an energy balance on the glass gives:

$$\alpha I_T = h_i \Delta T_{sol} + h_o \Delta T_{sol} \quad (6)$$

Solving this equation for ΔT_{sol} and replacing it in Eq. (5) results in:

$$\frac{\dot{Q}}{A} = U(T_0 - T_i) + \tau I_t + \frac{\alpha I_t h_i}{h_i + h_o} \quad (7)$$

Two rightmost terms can be combined as the total solar heat gain through single pane glazing,

$$\frac{\dot{Q}}{A} = U(T_0 - T_i) + SHGC \cdot I_t \quad (8)$$

Where,

$$SHGC = \tau + \frac{U}{h_o} \text{ with } U = \frac{h_i h_o}{h_i + h_o} \quad (9)$$

2.10.2. PMV and PPD

Fanger's thermal comfort model predicts occupant perception using the Predicted Mean Vote (PMV) and Predicted Percentage Dissatisfied (PPD) [38], which remain central to standards like ISO 7730 [39]. PMV depends on environmental conditions, activity level, and clothing insulation.

Fanger's conceptual heat balance equation is [40]:

$$H - E_d - S_w - E_{RE} - L = K = R + C \quad (10)$$

Fanger's PMV and PPD equations are as follows [40]:

$$\begin{aligned} PMV = & 0.303e^{-0.036M} + 0.028 \times [(M - W) - 3.05 \\ & \times 10^{-3} \{5733 - 6.99(M - W) - P_a\} - 0.42 \{(M - W) - 58.15\} \\ & - 1.7 \times 10^{-5} M(5867 - P_a) - 0.0014M(34 - T_a) - 3.96 \\ & \times 10^{-8} f_{cl} [(T_{cl} + 273)^4 - (T_r + 273)^4] - f_{cl} h_{cl} (T_{cl} - T_a)] \end{aligned} \quad (11)$$

Where,

$$\begin{aligned} T_{cl} = & 35.7 - 0.028(M - W) \\ & - 0.155I_{cl} [3.96 \times 10^{-8} f_{cl} [(T_{cl} + 273)^4 - (T_r + 273)^4] + h_{cl} (T_{cl} - T_a)] \end{aligned} \quad (12)$$

$$h_c = \max(2.38(t_{cl} + t_a)^{0.25}, 12.1\sqrt{v}) \quad (13)$$

$$\begin{aligned} f_{cl} = & 1.0 + 0.2I_{cl} \text{ for } I_{cl} \leq 0.5 \\ = & 1.05 + 0.1I_{cl} \text{ for } I_{cl} > 0.5 \end{aligned} \quad (14)$$

$$PPD = 100 - 95 \exp(-0.03353PMV^4 - 0.2179PMV^2) \quad (15)$$

Here, M is the metabolic rate (W/m²; in met units, 1 met = 58.15 W/m²), W is the external work (usually 0 for sedentary activities), P_a is the partial vapor pressure (Pa), T_a is the air temperature (°C), T_r is the mean radiant temperature (°C), f_{cl} is the clothing surface area factor (dimensionless), T_{cl} is the clothing surface temperature (°C), and h_c is the convective heat transfer coefficient (W/m²·K).

3. Results and discussions

3.1. Validation

Before analyzing the simulation results, a work of Garg et al. [41] was validated to check the compatibility of the software. Following the outlined methodology of the work by Garg et al., the WWR varied from 0% to 90% with an increment of 10%. The simulations were performed without incorporating the lighting control. Then the energy consumption data was documented and compared with the actual work. The trend of energy consumption was aligned closely with the work of Garg et al. Fig. 5 illustrates the comparison of energy consumption between the findings from Garg et al. and the results obtained from our work. The percentage of deviation for the validation was under 1%, which indicates a strong accuracy with the values reported by Garg et al. The

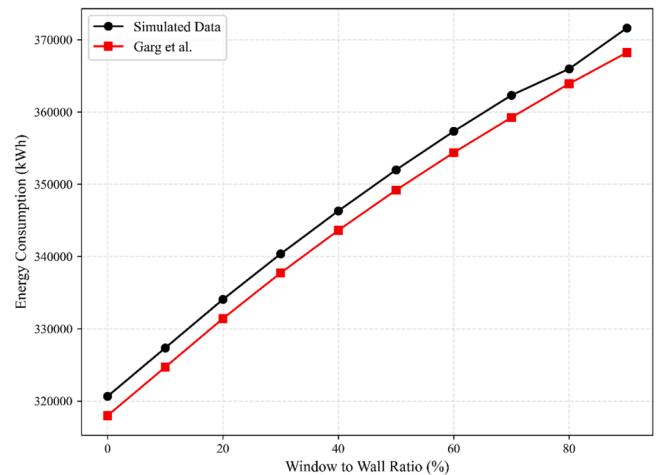


Fig. 5. Comparison between Garg et al. [41] and current investigation.

validation ensures that the software can accurately calculate the energy consumption of buildings in any given climate. The validation results demonstrate the relationship between increasing WWR and energy demand. In the main analysis, variations in window shape lead to differences in effective glazing area, which implicitly influence WWR. Therefore, the observed performance differences among window configurations can be partially interpreted in the context of WWR variation.

3.2. Effect of window configuration on solar gains

For calculating annual energy consumption, total solar gains through exterior windows need to be considered. The total solar gains through the exterior window have been plotted according to monthly data for each case. These plot helps in identifying seasonal variations in heat gain patterns across different window types and glass configurations. Such insights allow for evaluating which design strategies contribute to reduced cooling loads while maintaining adequate daylighting.

Fig. 6 presents the solar gains through exterior windows for each window configurations from January to December. Across all four window shapes, the solar heat gains through exterior windows show significant seasonal variation. Solar gains peaked in March to May in every case. And the lowest gain occurred in July, August and September. There is a secondary rise in October. Because the solar altitude decreases after September in hot and humid climate like Bangladesh. Since the windows are facing towards east and west direction, the lower angle hits the windows more directly and increases the gains higher compared to mid-summer. Also, in October the cloud cover is reduced, and the skies begin to clear which increases the direct solar radiation even though the ambient temperature is slightly lower. As a result, the solar gains further increase.

Across all the glazing types, clear glass showing the highest solar gains because clear glass transmits most solar radiation, which directly correlates the SHGC of clear glass of 0.819. Bay shaped windows with clear glass results in the greatest solar gains among all other configurations. On the other hand, the minimum solar gains are for grey glass, especially Palladian shaped windows with grey glass provides the lowest solar gains a total of 311,385.58 kWh annually.

Clear glass consequently shows higher heat gains, where tinted glass shows a significant reduction in solar gains. The values for tinted glass are almost clustered together where grey glass showing the minimum solar gains for all window shapes. Bay windows are responsible for the highest solar gains because of their larger surface area of 4.256 m², whereas Palladian shaped windows reduce solar gains by 66.10% compared to the bay clear glass windows, as the area of a Palladian window is 2.70 m².

3.3. Effect of window configurations on energy consumption

3.3.1. Annual cooling energy consumption

Cooling energy demand constituted the largest portion of energy consumption for hot and humid climate of Jashore. The shape of the windows and the SHGC of the glasses are the primary determinant of cooling load. Fig. 7 demonstrates the annual cooling energy demand patterns for all five glasses for rectangular windows. Clear glass requires higher cooling demand than the other tinted glasses. The annual cooling energy demand for the baseline case is 59,920.943 kWh. For Arched and Palladian shaped windows with clear glass the cooling energy demand decreased to 54,900.607 kWh and 54,114.457 kWh, respectively. But for the tinted glass for these two shaped windows the annual cooling energy demand reduced significantly. Palladian green glass configuration consumes the lowest amount of cooling electricity among all the configurations. The annual cooling energy demand for Palladian green glass configuration is reduced by 8101.176 kWh compared to the baseline. Bay shaped windows are responsible for highest amount of annual cooling load across all four shapes. The excess amount of annual cooling load is due to the large surface area of the Bay windows. The cooling demand remains higher for clear glass, whereas the tinted glasses show almost identical performance. The Bay shaped windows with clear glass consume 66,597.388 kWh for cooling, which is the highest among all the cases. Cooling energy results validate the effectiveness of tinted glasses in reducing thermal loads. The cooling energy patterns directly correlate with solar gains analysis. The reductions directly decrease the HVAC operational costs and reduce peak electrical demand during high temperature.

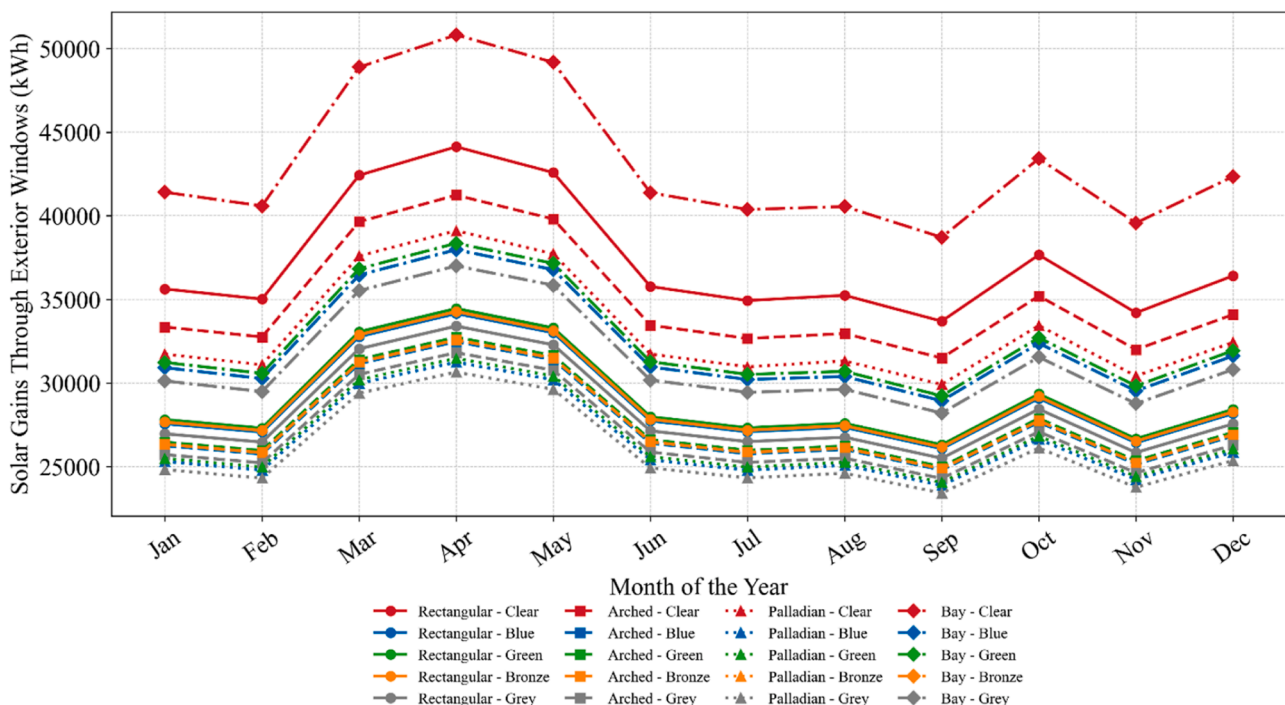


Fig. 6. Solar gains through exterior windows for each window configurations.

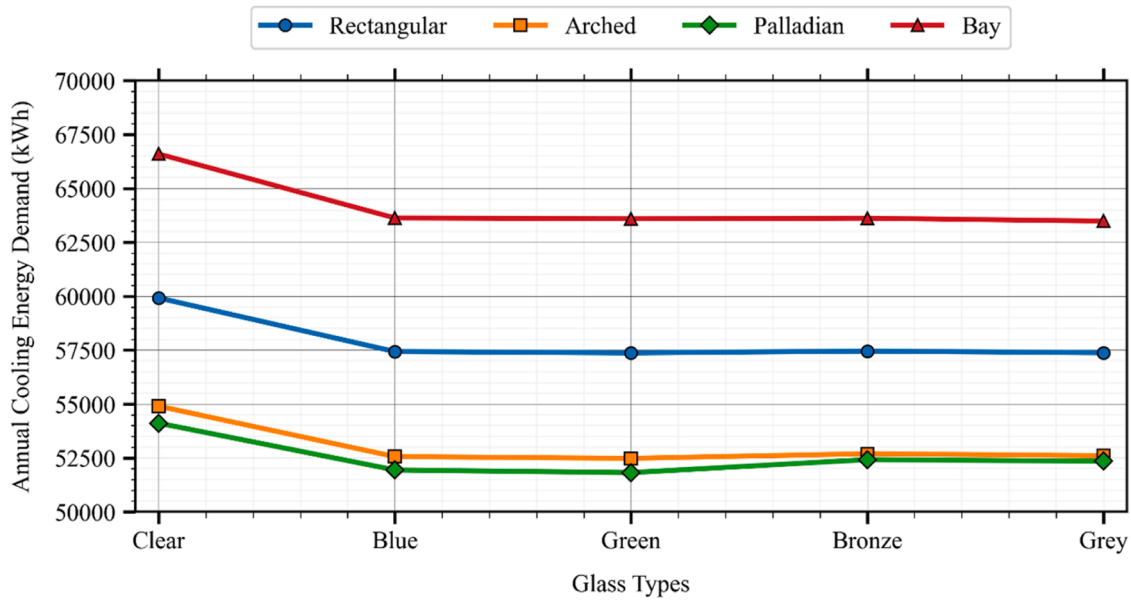


Fig. 7. Annual cooling energy demand through different configurations of windows.

3.3.2. Lighting energy consumption

There is an inverse relation between lighting energy consumption and daylighting availability. The lighting energy analysis of different types of glass for different window shapes is presented in Fig. 8. Clear glass shows the lowest amount of lighting energy consumption annually. Green glass consumes slightly higher lighting energy than the clear glass. But grey glass consumes the highest amount of lighting energy because of its lower visible transmittance. Lighting energy consumption is nearly the same for both blue and bronze glass. However, the bronze glass exhibits the lighting energy consumption marginally. In terms of window shape, Palladian windows show slightly higher lighting energy consumption than other window shapes across all glazing configurations. Due to their reduced glazing area, Palladian windows require a greater amount of lighting energy.

3.3.3. Total annual energy consumption

The total annual energy consumption is mainly influenced by cooling demand, lighting energy consumption and equipment gains. Fig. 9 illustrates the total annual energy consumption of the simulated cases. The results in Fig. 9 indicate that clear glass consistently consumes the

highest amount of energy annually across all the shapes. However, the green glass consumes the lowest. Bay windows with clear glass exhibit the highest total energy consumption at 142,573.386 kWh. This is due to increased cooling loads resulting from high solar gains through exterior windows because of large surface area and less efficient daylight distribution increasing the need of using artificial lighting. Bay shape geometry is the least energy efficient with an average of approximately 8% higher annual energy across five types of glasses than the baseline. Where Bay windows are less energy efficient, Arched and Palladian windows achieve superior result. With the Palladian green glass combination, the lowest energy consumption is found 119,468.311 kWh, reduced by 11,197.611 kWh annually compared to rectangular clear glass configuration. The Arched green glass combination consumes the second lowest annual energy which makes it the second highest energy efficient configuration of reduction in 11,038.104 kWh annually compared to the baseline. The Palladian blue and Arched blue configurations demonstrate a balanced performance as well. But the amount of daylighting penetration is reduced due to the lower value of visible transmittance of blue glass. Green glass demonstrates the best overall performance due to its balanced SHGC and visible light transmission.

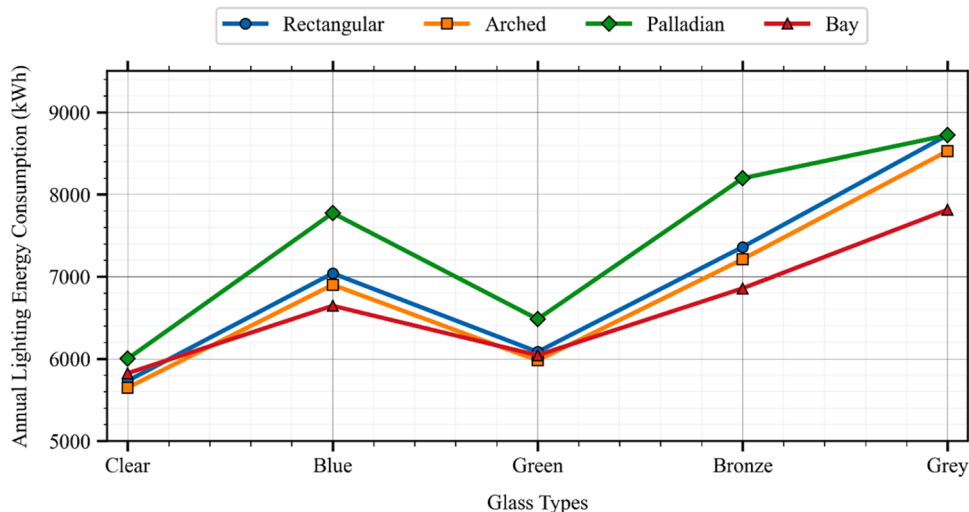


Fig. 8. Annual lighting energy consumption through different configurations of windows.

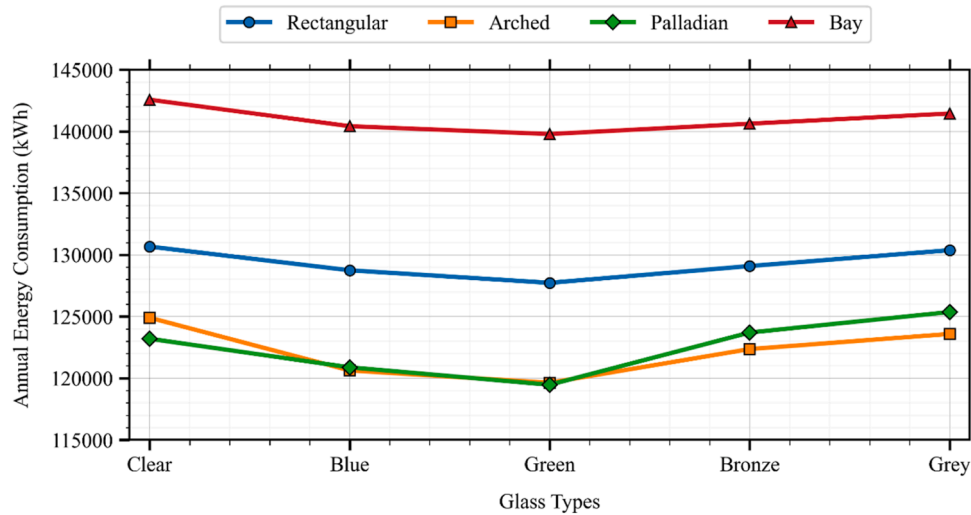
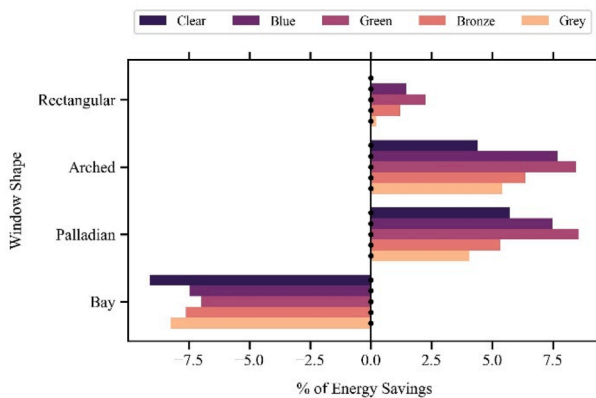
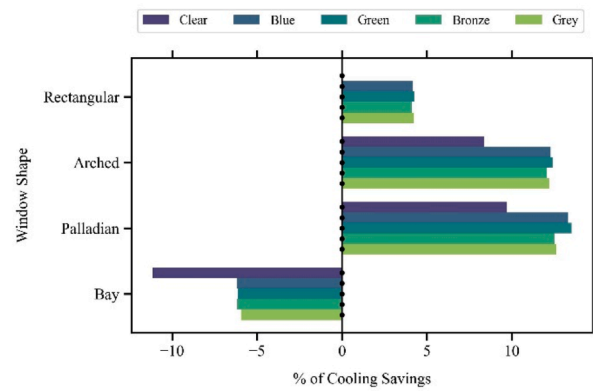


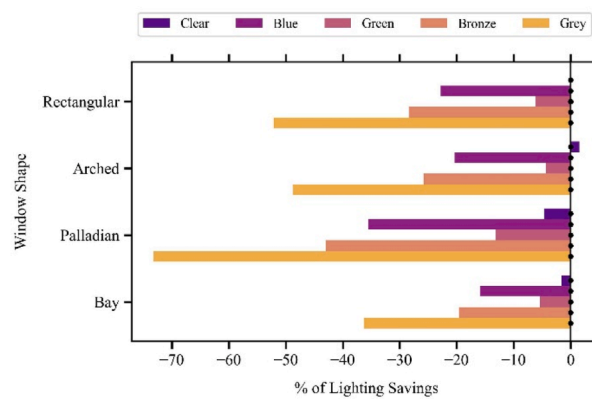
Fig. 9. Annual energy consumption for different configurations of windows.



(a)



(b)



(c)

Fig. 10. Percentage variation in (a) total, (b) cooling and (c) lighting energy consumption savings, for different window–glass configurations relative to the baseline.

Fig. 10 summarizes the comparative analysis on percentage change in total energy use, lighting energy consumption and cooling energy demand.

The results in Fig. 10(a) indicate that the impact of window design is not consistent for all energy components. For overall energy, Palladian

green and Arched green glass configuration consistently produced the maximum savings to 8.57% and 8.44% respectively, while Palladian and Arched windows with blue glass also performed well. On the other hand, Bay windows always showed negative values, clear glass increases the total energy use by 9.11% compared to the baseline. Cooling energy

performance showed a similar trend in Fig. 10(b). With tinted glass configurations delivering significant savings in annual cooling energy consumption. Palladian windows with green glass achieve the maximum cooling energy savings of about 13.51%, closely followed by Palladian-blue glass configuration. Bay windows again performed the worst, with clear glass resulting in a 11.14% increase in cooling energy demand compared to the baseline. In Fig. 10(c), lighting energy consumption shows substantial penalties with tinted glazing. Grey and bronze glasses demonstrated the largest reductions in lighting efficiency, with values as low as 73.24% in Palladian windows with grey glass combination. Only Arched-clear glass configuration showed a marginally positive improvement by 1.53%.

These results of Fig. 10 demonstrate that there is no single window-glass configuration which gives the best result for all energy end-use categories. Tinted glass reduces cooling energy demand, but it also increases lighting energy consumption. Palladian green glass configuration provides greatest savings in cooling load and overall energy usage and with relatively minor increase in lighting requirement. This configuration is considered as the most energy efficient configuration. In contrast, Bay windows have a low performance factor and are not ideal for thermal efficient buildings in hot and humid climate like Bangladesh. Bay clear glass configuration is identified as the worst compared to baseline. However, the practical application of window design is not solely governed by energy performance. Although bay windows exhibit higher energy consumption due to increased solar exposure and larger effective glazing area, they remain widely used in Bangladesh, especially in the residential buildings. This is primarily due to their architectural appeal, improved outdoor connectivity, and potential to enhance natural ventilation. This observation highlights an important trade-off between energy efficiency and architectural functionality. Therefore, while bay windows may not be optimal from a thermal performance perspective, their continued use in real-world applications is influenced by multiple design considerations beyond energy performance alone.

The energy performance of the Palladian window appears better partly because it has less glazing area. Since smaller window areas typically allow less solar heat to enter, both the shape and the size of the glazing play a role here. If the window area were increased, this advantage would likely not be as strong.

3.4. Thermal comfort performance

3.4.1. PMV and PPD analysis

Air-conditioned rooms typically provide a more stable and controllable thermal environment, making it easier and more meaningful to evaluate PMV accurately [42]. PMV analysis in AC rooms is highly relevant for assessing occupant comfort levels and optimizing air conditioning performance [43]. The thermal comfort assessment for a fourth-floor room named “ESE 506” was conducted using monthly average PMV values across twenty cases. The results are illustrated in Fig. 11 as a heatmap, which visually emphasizes the variation in thermal

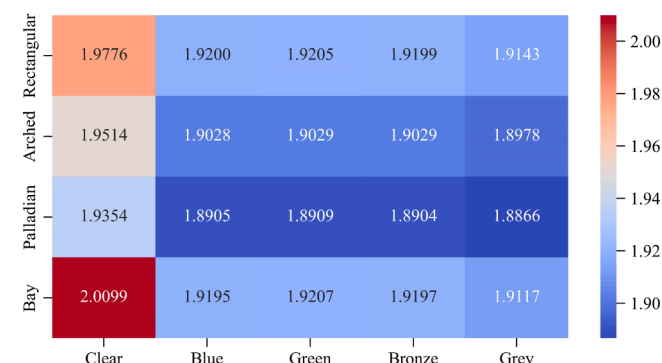


Fig. 11. PMV values for different window design strategies in room ESE 506.

comfort between the different design strategies.

Palladian shaped windows are identified as the most favorable geometry, consistently achieving the lowest PMV values ranges from 1.88 to 1.93. The reduced surface area and passively moderate solar penetration because of the architectural design of Palladian windows highlight improving thermal comfort. Bay windows produce the highest PMV values, with Bay-clear glass configuration peaking at 2.01. The increased glazed area and higher exposure of direct solar radiation elevate window heat gain. The PMV values for rectangular shaped windows also showed relatively high value ranges from 1.91 to 1.97, though slightly lower than Bay windows. The conventional flat design without shading remains prone to overheating. Arched windows display an improvement over rectangular window. The curved top reduces the solar heat gain. The incorporation of simple external shading devices, such as overhangs or louvers, could effectively reduce solar radiation and improve thermal performance, potentially offering a cost-effective alternative. However, complex window geometries such as Palladian or arched designs are often selected not only for performance but also for their architectural aesthetics and visual character.

Clear glass is consistently associated with the highest PMV values, confirming its poor thermal performance control. Tinted glass types provide reductions in PMV. Although the differences were minor, the effect was consistent and measurable. Specifically, grey glass offers a better performance across all window shapes. The darker tones with the lower SHGC and lower visible transmittance of grey glass contribute positively to reducing indoor overheating. Although the PMV values remain within the warm comfort category, they do not reach the neutral zone (0.0 to ± 0.5). Continuous occupancy in such conditions leads to discomfort, particularly during peak summer months.

Fig. 12 shows the relationship between PPD and PMV, plotted based on the monthly average data for ESE 506's thermal environment using Palladian green glass configuration.

The results in Fig. 12 present the critical relationship between the thermal comfort indicators PMV and PPD. There is a positive correlation between PMV and PPD, as the PMV value increases, the PPD also increases. This provides insights into how changes in the room's thermal environment can affect occupant satisfaction.

3.4.2. Operative temperature analysis

An operative temperature analysis was conducted for selected cases (baseline, best, and worst configurations) to assess thermal conditions under naturally ventilated scenarios. This assessment is based on locally established operative temperature ranges for classrooms in Bangladesh, as reported in the literature. According to a study of Talukder et al. [44], the operative temperature ranges from 24.7 to 33.5 °C in summer season for a classroom in Bangladesh to ensure students' comfort with the thermal environment. The hourly distribution of operative temperature, relative humidity and humidity ratio over a year for the building is plotted in a psychrometric chart for these configurations. To plot the hourly data in psychrometric chart, the operative temperature range was set as the findings from Talukder et al.

Fig. 13 illustrates the annual hourly distribution of indoor operative temperature, relative humidity, and humidity ratio for naturally ventilated classrooms. Fig. 13(a) represents the psychrometric chart with using Rectangular clear glass configuration, considered as the baseline. Fig. 13(b) presents the temperature distribution with the rooms fitted with Palladian green glass windows, identified as the best-performing configuration. And Fig. 13(c) presents the indoor thermal environment for classrooms with Bay clear glass windows, which showed the worst performance among all cases. The psychrometric chart (Fig. 13(b)) shows a more concentrated distribution of data points within a narrower range of operative temperature and humidity ratio compared to the baseline case presented in Fig. 13(a), indicating relatively improved indoor thermal conditions. This behavior can be attributed to the reduced solar heat gain associated with green glass, which helps moderate indoor temperature fluctuations. In contrast, Fig. 13(c) reveals a

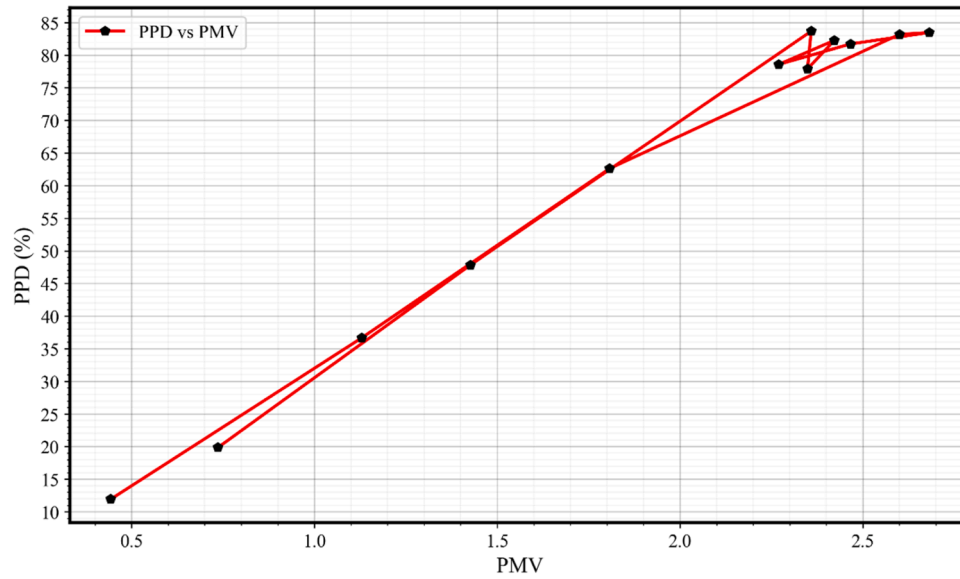
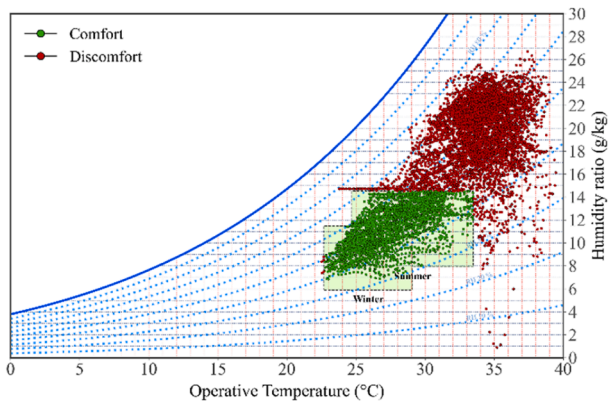
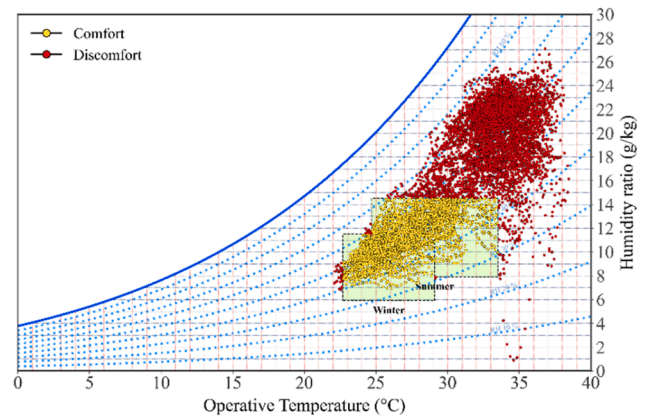


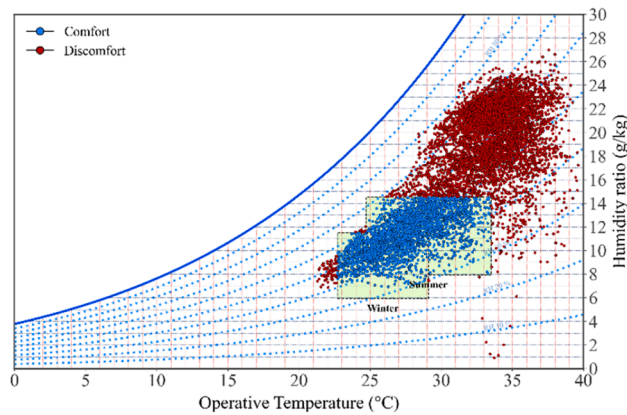
Fig. 12. PPD vs PMV analysis in room ESE 506 using Palladian green glass configuration.



(a)



(b)



(c)

Fig. 13. Annual operative temperature distribution for (a) Rectangular-clear glass configuration, (b) Palladian-green glass configuration, and (c) Bay-clear glass configuration.

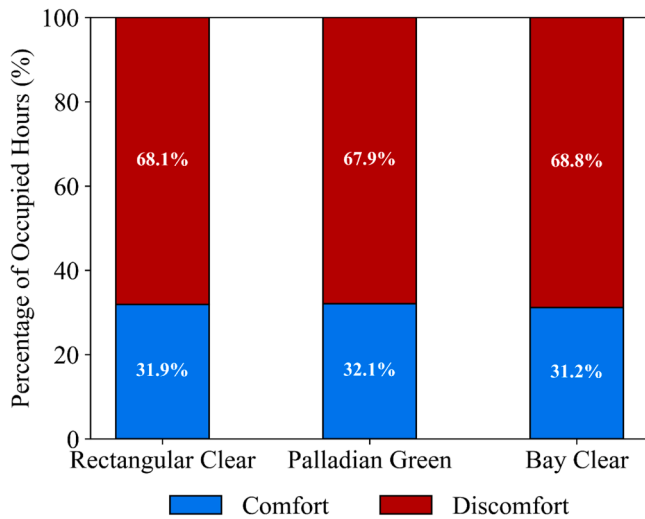


Fig. 14. Comparison of thermal comfort and discomfort percentages for three window configurations.

wider spread of operative temperature and humidity ratio, with a greater number of hours extending beyond the acceptable range, indicating less stable indoor conditions.

The stacked bar chart shows in Fig. 14 gives a clear visualization of the proportion of time (annual occupied hours) spent within comfort zones and discomfort zones which is defined by thresholds for operative temperature and humidity ratio.

As seen on Fig. 14, the percentage of thermal comfort values are 31.9%, 32.1% and 31.2% for Rectangular clear, Palladian green and Bay clear glass configurations, respectively. The Palladian-green glass configuration demonstrates a slight improvement, showing a higher percentage of occupied hours within the acceptable operative temperature range based on locally established comfort limits for classroom environments, indicating relatively improved indoor thermal conditions compared to other configurations. This demonstrates the positive impact of the Palladian green glass configuration in maintaining occupant comfort in naturally ventilated classrooms through passive strategies. As the average temperature in Bangladesh is generally high, maintaining a comfortable thermal environment is difficult only by retrofitting the window geometry and glazing type. However, the Palladian shaped

window shows consistent results in every possible outcome where it can be clearly said that Bay shaped window is not suitable for hot and humid climate like Bangladesh.

3.5. Daylighting performance evaluation

The visual comfort performance of the building was evaluated through detailed illuminance analysis across different floors, with particular emphasis on the ground floor and fourth floor (top floor) comparison. Table 8 summarizes the daylighting performance of different window shapes combined with various glazing types in terms of maximum and average indoor illuminance, average daylighting factor, visual comfort, and excessive glare percentages.

Bay windows achieve highest amount of illuminance levels and daylighting factor among all other configurations, resulting in improving visual comfort. However, this advantage accompanied by a comparatively higher risk of excessive glare, especially for clear glazing. Palladian and arched windows demonstrate moderate daylight availability, with clear and green glazing providing a balanced trade-off between adequate illuminance and controlled glare. On the other hand, bronze and grey glazing shows reduced daylighting as well as visual comfort across all the configurations. Overall, the results highlight the influence of both window shape and glazing on indoor daylighting availability, visual comfort and excessive glare control.

3.5.1. Spatial illuminance distribution patterns

Since the Palladian window with green glass achieved the highest energy savings among all configurations, the spatial distribution of indoor illuminance on both the ground floor and the fourth floor was further analyzed using this configuration. This analysis provides deeper insight into the daylighting performance and uniformity of illuminance across different floor levels.

Fig. 15(a) and 15(b) show the daylighting contour maps using Palladian-green glass combination for the ground floor and fourth floor, respectively. It is observed from Fig. 15(a) that, most of the spaces are dominated by lower illuminance values but near the open space the illuminance values are higher. The vertical thin rectangles represent the shadow cast by the external pillars of the western side of the building. In Fig. 15(b), there is a decent daylighting distribution in fourth floor. A detailed analysis of east and west-facing rooms provides a clear understanding of the daylight distribution.

Fig. 16(a) and 16(b) highlight the daylighting contour of east and west facing zones using Palladian-green glass configuration for ground

Table 8

Comparison of daylighting performance for different window configurations.

Window Shape	Glazing Type	Max Illuminance (lux)	Avg Illuminance (lux)	Avg Daylighting Factor (%)	Visual Comfort (%)	Excessive Glare (%)
Rectangular	Clear	3418.86	1709.43	1.815	17.31	3.61
	Blue	3445.76	1722.88	1.362	11.17	2.12
	Green	3459.67	1729.835	1.619	15.03	2.83
	Grey	3450.78	1725.39	1.17	7.58	1.95
	Bronze	3463.06	1731.53	1.312	10.29	1.99
Arched	Clear	3511.04	1755.52	1.564	13.98	2.17
	Blue	3455.01	1727.505	1.209	8.51	1.97
	Green	3441.02	1720.51	1.41	11.96	1.97
	Grey	3421.39	1710.695	1.059	5.61	1.95
	Bronze	3467.72	1733.76	1.17	7.69	1.98
Palladian	Clear	3528.83	1764.415	1.521	13.11	3.07
	Blue	3473.37	1736.685	1.184	8.56	2.10
	Green	3500.82	1750.41	1.376	11.58	2.36
	Grey	3450.77	1725.385	1.038	6.75	1.96
	Bronze	3480.14	1740.07	1.145	8.02	2.09
Bay	Clear	3710.56	1855.28	2.219	19.14	5.22
	Blue	3695.12	1847.56	1.564	11.30	3.24
	Green	3718.18	1859.09	1.883	15.54	4.39
	Grey	3682.97	1841.485	1.329	7.90	2.73
	Bronze	3716.54	1858.27	1.502	10.40	3.08

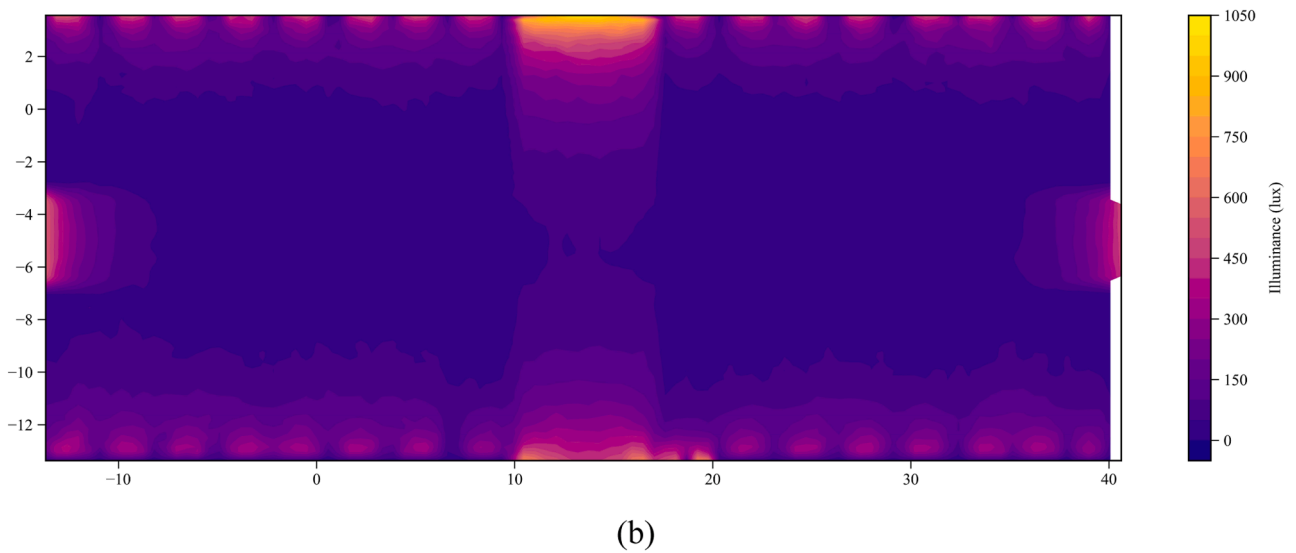
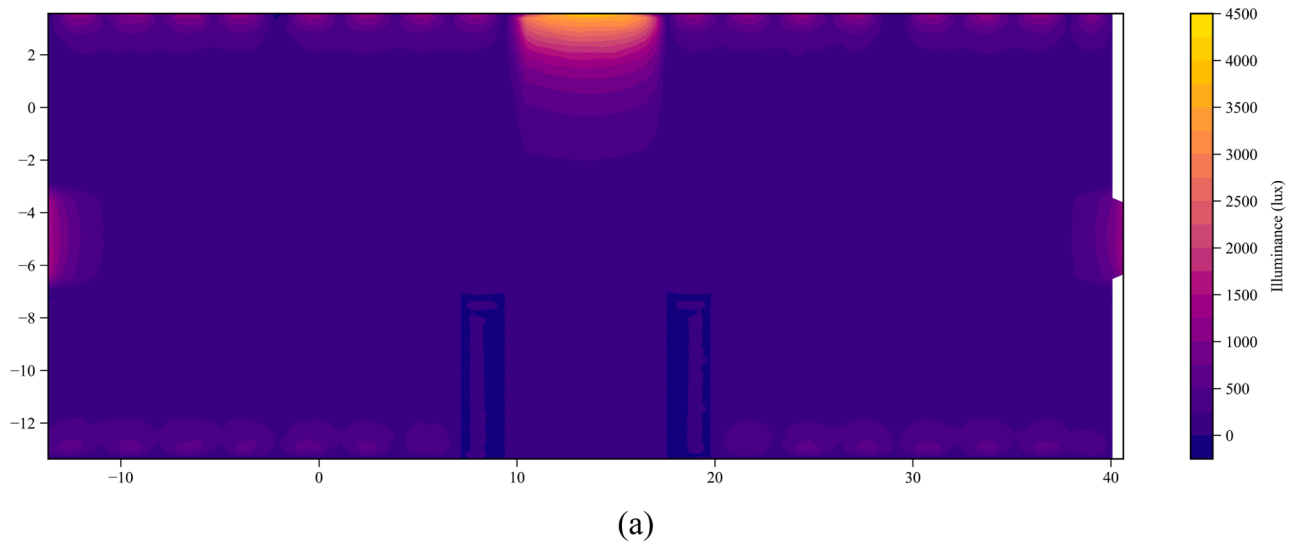


Fig. 15. Daylighting contour maps using Palladian-green glass configuration: (a) ground floor, (b) fourth floor.

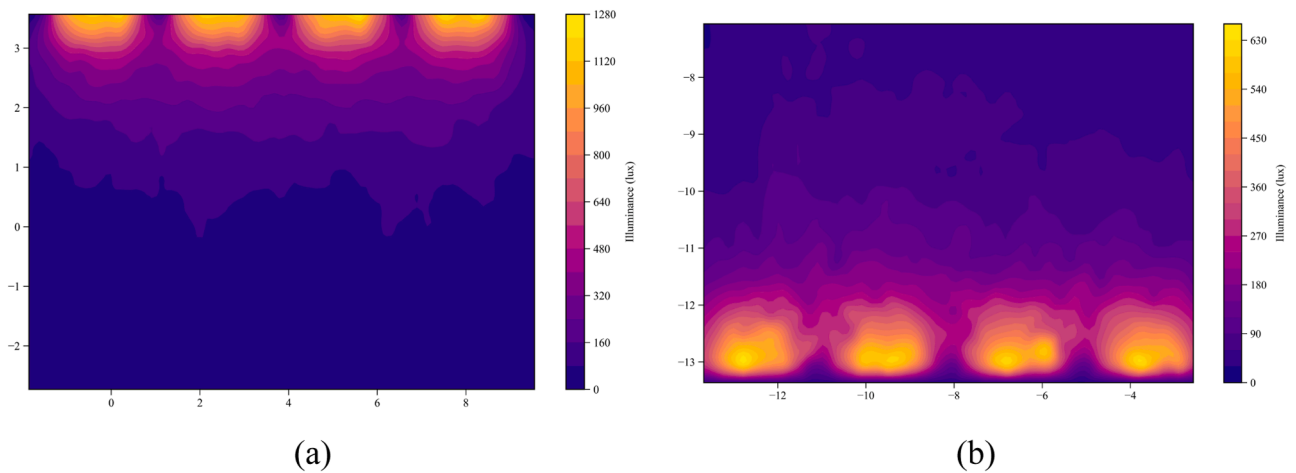


Fig. 16. Daylighting contour maps of the ground floor using Palladian-green glass configuration: (a) east-facing room, (b) west-facing room.

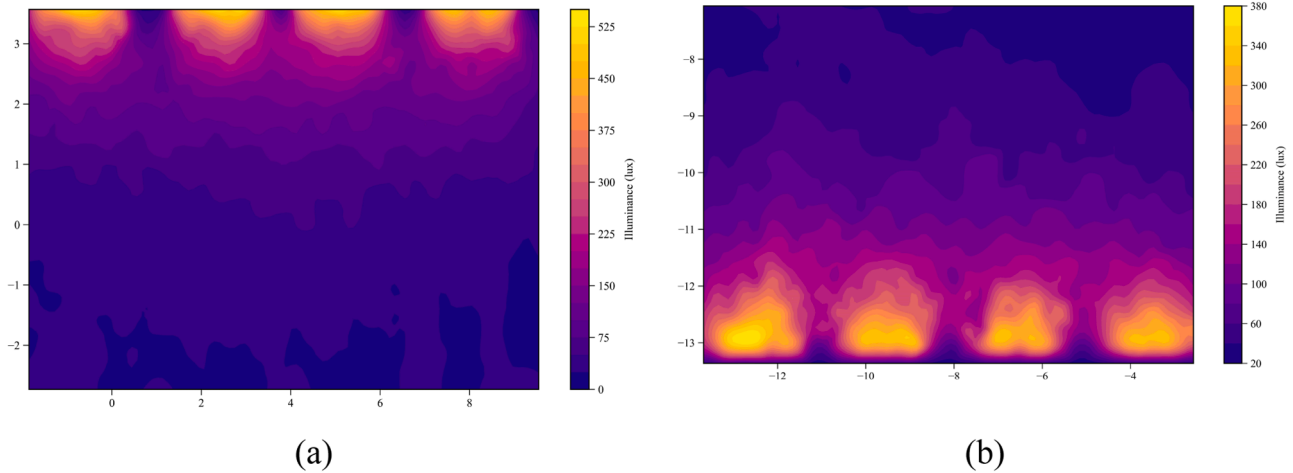


Fig. 17. Daylighting contour maps of the fourth floor using Palladian-green glass configuration: (a) east-facing room, (b) west-facing room.

floor. And Fig. 17(a) and 17(b) shows the daylighting contour of east and west facing zones using Palladian-green glass configuration for fourth floor.

East-facing room receives direct morning sunlight, resulting in higher and concentrated illuminance near windows, while west-facing rooms obtain more diffuse, afternoon light with generally lower peaks. Overall, while Palladian-green glass can modify window performance, both orientation and floor height significantly influence indoor daylight distribution.

In contrast, the afternoon sun is lower in altitude than the morning sun in Bangladesh. This lower oblique angle results in penetrating direct beam radiations into the west-facing windows, which causes bright glare near the window surface while leaving the adjacent area relatively dim.

3.5.2. Visual comfort analysis

The percentage values of visual comfort hours, under-lit hours and excessive glare are plotted for the ground floor and fourth floor for the best performing (Palladian-green) configuration. Fig. 18 illustrates the glare percentages for the Palladian-green configuration on the ground floor (a) and the fourth floor (b).

For Palladian-green glass configuration, the rates of visual comfort are 8.7% and 13.3% respectively for ground floor and fourth floor. But the rates of excessive glare are only 0.3% and 4.6% for ground floor and fourth floor respectively. While slightly compromising visual comfort, the Palladian-green glass configuration effectively lowers excessive glare.

The Palladian green configuration demonstrates the lowest cooling energy requirement; however, it is associated with a higher proportion of under-lit conditions. This indicates a trade-off between thermal performance and daylight availability. The reduced daylight levels are

likely due to the orientation of the selected building and the smaller glazing area, which restricts light penetration into interior spaces. Consequently, artificial lighting demand increases, leading to higher lighting energy use. Nevertheless, the total annual energy consumption remains lower than other configurations, as the decrease in cooling energy outweighs the increase in lighting demand.

Although its visual comfort performance is slightly lower than some other configurations, it has a relatively low glare level (2.36%), indicating better visual conditions in terms of glare control. This means the design balances reduced daylight by limiting excessive glare while maintaining energy efficiency. However, reduced daylight may still affect comfort and productivity, particularly in educational environments. Therefore, while the Palladian green configuration offers significant energy-saving potential, its practical application should consider both energy performance and indoor environmental quality to achieve a balanced design outcome.

3.5.3. Daylighting factor analysis

Fig. 19. represents how Daylighting Factor (DF) varies with the distance from the window from room ESE 506 in fourth floor for Palladian-green glass configuration. There is a risk of high Daylighting Factor (DF) near the window surface. But as the distance from the window increases the Daylighting factor decreases. The value of Daylighting Factor ranges approximately from 0.5 to 3.5.

The reduced daylighting factor near the window is relatively low due to the shadow cast by the window itself, and it remains low for a short distance. After that, the DF peaks and then decreases uniformly with increasing distance from the window.

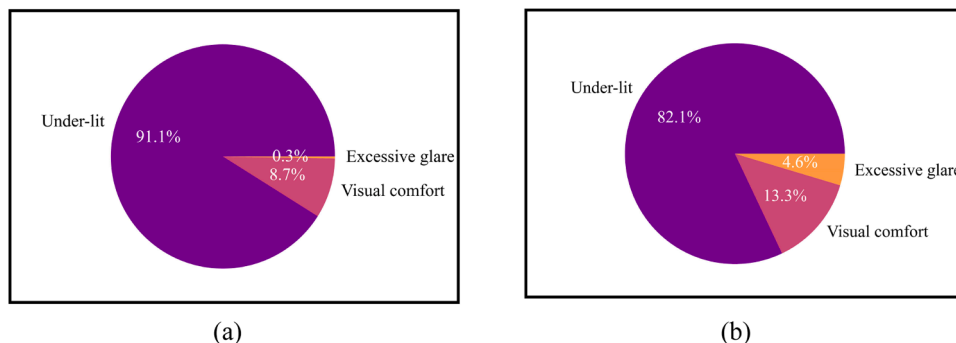


Fig. 18. Glare percentage for Palladian-green glass configuration (a) ground floor, (b) fourth floor.

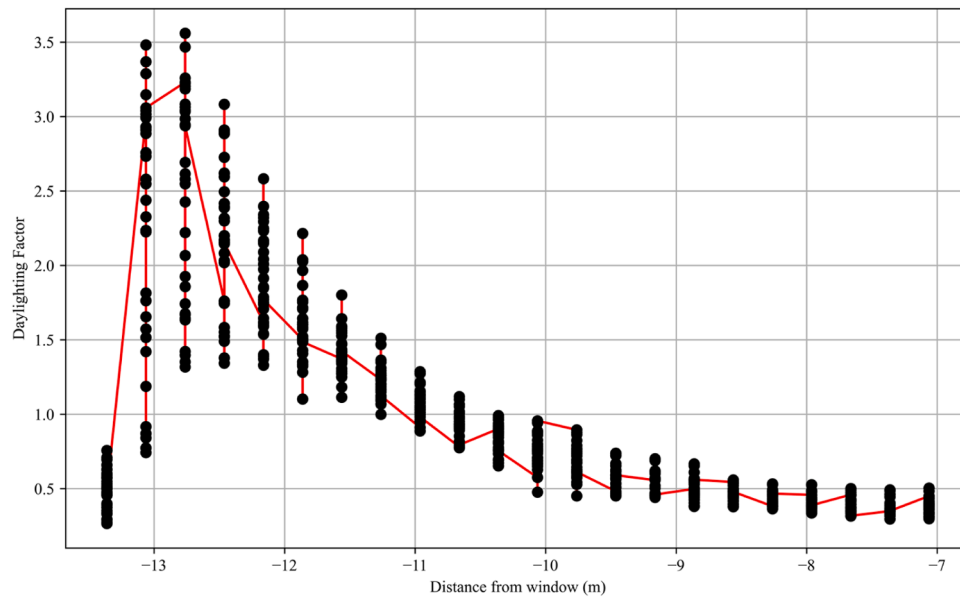


Fig. 19. Daylighting factor vs distance from the window.

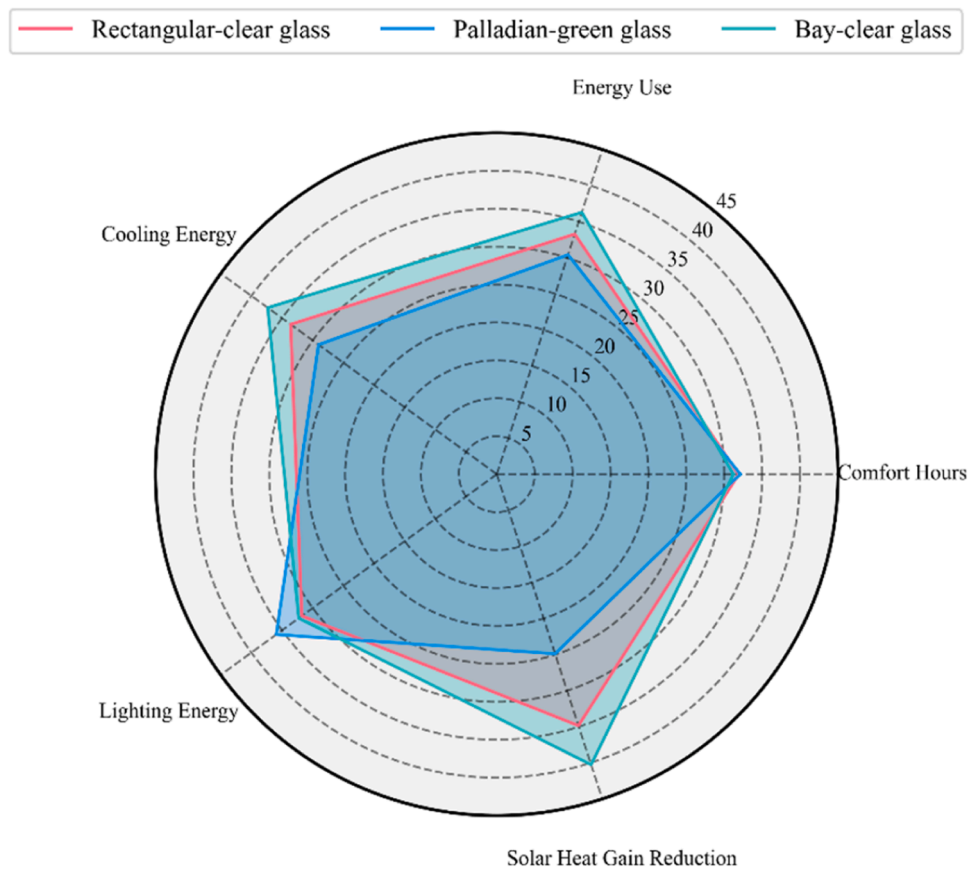


Fig. 20. Comparative analysis between three different window design configurations.

3.6. Comparative performance analysis

In Fig. 20, a radar chart illustrates the comparative analysis of three different window configurations: the baseline case of rectangular windows with clear glass, the best-performing case of Palladian windows with green glass, and the worst-performing case of Bay windows with

clear glass. The comparison is based on five different indicators: Energy Use, Lighting Energy Consumption, Cooling Energy Demand, Comfort Hours and Solar Gains Through Exterior Windows.

The window design performance analysis demonstrates that window design significantly impacts on building energy consumption, lighting energy consumption, cooling energy demand, thermal comfort and

daylighting penetration. The result reveals that Palladian-green glass configuration is the most energy efficient option, reducing the highest amount of energy consumption. Palladian-green glass configuration also offers a balance performance to ensure thermal and visual comfort. On the other hand, clear-glazed Bay windows were observed to have the least energy performance due to high cooling loads for summer seasons and lower thermal comfort. These results highlights that the window geometry as well the glazing material are important when we consider the overall building energy performance and indoor environmental quality.

4. Conclusion

The study investigates the influence of window design strategies in building energy consumption, indoor environmental quality i.e. thermal comfort and visual comfort in hot-humid climate through numerical simulation in DesignBuilder software. Through a structured exploration on different window shapes combined with clear and tinted (Blue, Green, Bronze, Grey) glass across twenty simulation cases, several practical and conclusive insights were obtained. Window geometry plays a significant role in determining solar gains, overall energy efficiency and daylighting uniformity. The key findings from this study are:

- Clear glass maximizes the solar gains through exterior windows. Despite increasing daylighting penetration and reducing annual lighting energy consumption, it led to higher cooling load due to higher solar gains and excessive glare.
- Tinted glass reduces the cooling loads and glare while still transmitting adequate amount of daylighting. The tinted glass achieves a reduction in cooling energy demand ranging between 12% and 13% for both Arched and Palladian shaped window.
- While performing a balanced daylighting, green glass reduces the highest amount of cooling demand as well as overall energy consumption. The Palladian-green glass shows highest amount of reduction in energy consumption by 8.56% compared to the baseline.
- Bay windows consistently showed the highest amount of solar gain as well as cooling energy demand. Though Bay-clear glass configuration gives the highest rate of visual comfort, the increased glare throughout the day rises the heat gains because of their orientation toward multiple solar angles.
- Palladian windows emerge as the most favorable window shape. Their segmented openings limit the excessive solar gains while allowing sufficient daylighting distribution. The Palladian-green glass configuration shows a monthly average PMV of 1.89, which means a slightly warm condition but better than the other options. From the operative temperature analysis, this configuration provides 32.1% comfort hours in a year. And through visual comfort analysis, it shows fewer visual comfort hours but the lowest excessive glare, which helps reduce heat gain.

These results suggest that selecting appropriate window geometry combined with moderate tinting is an effective passive strategy to enhance energy performance in institutional buildings in Bangladesh. Future study should consider incorporating machine learning to predict building energy performance and daylighting availability from different data driven window design strategies.

5. Future research direction

This study evaluates the effects of window geometry and glazing type on building energy performance and indoor thermal and visual comfort, without incorporating external shading elements. External shading devices such as overhangs, fins, louvers, etc. can significantly reduce solar heat gain and improve thermal comfort. Future work should explore the combined influence of window configurations and shading strategies to

achieve more efficient designs.

In addition, to determine the feasibility of different design options, economic factors such as initial cost, maintenance, and lifecycle performance should be considered. While complex window geometries may offer performance benefits, simpler solutions like shading devices may prove more cost-effective in practical applications.

Moreover, advanced optimization techniques, including machine learning and data-driven approaches, can be employed to explore extensive design possibilities and identify optimal combinations under different climatic conditions.

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Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT-4 in order to improve language and readability, with caution. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

CRediT authorship contribution statement

Tahmid Talukder: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Md. Mahmudul Alam:** Writing – review & editing, Writing – original draft, Supervision, Software, Methodology, Formal analysis, Conceptualization. **Pronab Debnath:** Writing – review & editing, Writing – original draft, Methodology, Data curation. **MAH Shovon:** Writing – review & editing, Writing – original draft, Data curation. **Fahim Tanvir:** Visualization, Formal analysis. **Nazmush Sakib:** Writing – review & editing, Writing – original draft. **Ahbab Faiyaj Haque:** Writing – review & editing, Writing – original draft. **Md. Hasan Ali:** Writing – review & editing, Writing – original draft, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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